

Pizza Sales Analysis Using SQL

Data Analytics Portfolio Project
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Project Overview

Objective:

- Analyze pizza sales using SQL.
- Answer 13 business questions.
- Generate actionable business insights.

Dataset:

Orders, Order Details, Pizzas, Pizza Types.

Dataset Relationship

orders → order_details → pizzas → pizza_types

Primary Keys:

- order_id
- pizza_id
- pizza_type_id

Used joins to connect transactional and product data.

SQL Skills Demonstrated

- JOINS
- Aggregate Functions
- GROUP BY
- ORDER BY
- Window Functions
- Subqueries
- CTEs
- Ranking Functions
- Business Analysis

Analysis 1: Total Orders

Retrieve the total number of orders placed.

```
select count(order_id) as total_orders from orders;
```

Result Grid	
	total_orders
▶	21350

Analysis 2: Total Revenue

Calculate the total revenue generated from pizza sales.

```
SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
          2) AS Total_revenue
FROM
    order_details
    JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

Result Grid	
	Total_revenue
▶	817860.05

Analysis 3: Highest-priced Pizza

Identify the highest-priced pizza.

```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    INNER JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Result Grid				 Filter Row
	name	price		
▶	The Greek Pizza	35.95		

Analysis 4: Most Common Pizza Size

Identify the most common pizza size ordered.

```
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
    JOIN
        order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size      -- group by is must for the aggregate value count
ORDER BY order_count DESC;
```

Result Grid				 Filter Results
	size	order_count		
▶	L	18526		
	M	15385		
	S	14137		
	XL	544		
	XXL	28		

Analysis 5: Top 5 Ordered Pizza Types

List the top 5 most ordered pizza types along with their quantities.

```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Result Grid			Filter Rows:
	name	quantity	
▶	The Classic Deluxe Pizza	2453	
	The Barbecue Chicken Pizza	2432	
	The Hawaiian Pizza	2422	
	The Pepperoni Pizza	2418	
	The Thai Chicken Pizza	2371	

Analysis 6: Category-wise Quantity

Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

Result Grid			Filter
	category	quantity	
▶	Classic	14888	
	Supreme	11987	
	Veggie	11649	
	Chicken	11050	

Analysis 7: Orders by Hour

Determine the distribution of orders by hour of the day.

```
select hour(order_time) as hour,  
count(order_id) as order_count  
from orders  
group by hour(order_time);
```

Result Grid					Filter
	hour	order_count			
▶	11	1231			
	12	2520			
	13	2455			
	14	1472			
	15	1468			
	16	1920			
	17	2336			
	18	2399			
Result 1			×		

Analysis 8: Category Distribution

Join relevant tables to find the category-wise distribution of pizzas.

```
select category, count(name)
from pizza_types group by category;
```

Result Grid			Filter Rows:
	category	count(name)	
▶	Chicken	6	
	Classic	8	
	Supreme	9	
	Veggie	9	

Analysis 9: Average Pizzas per Day

Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT      -- query or outer query
  ROUND(AVG(quantity), 0) AS avg_pizza_ordered_per_day
FROM
  (SELECT      -- subquery or inner query or derived query
    orders.order_date, SUM(order_details.quantity) AS quantity
  FROM
    orders
  JOIN order_details ON orders.order_id = order_details.order_id
  GROUP BY orders.order_date) AS order_quantity;
```

Result Grid				Filter Rows:
	avg_pizza_ordered_per_day			
▶	138			

Analysis 10: Top 3 by Revenue

Determine the top 3 most ordered pizza types based on revenue.

```
select pizza_types.name,  
sum(order_details.quantity * pizzas.price) as revenue  
from pizza_types join pizzas  
on pizzas.pizza_type_id = pizza_types.pizza_type_id  
join order_details  
on order_details.pizza_id = pizzas.pizza_id  
group by pizza_types.name order by revenue desc limit 3;
```

Result Grid				 Filter Rows:
	name	revenue		
▶	The Thai Chicken Pizza	43434.25		
	The Barbecue Chicken Pizza	42768		
	The California Chicken Pizza	41409.5		

Analysis 11: Revenue Contribution

Calculate the percentage contribution of each pizza type to total revenue.

```
select pizza_types.category,  
       round(sum(order_details.quantity * pizzas.price) / (select  
         round(sum(order_details.quantity * pizzas.price),2)  
         as total_sales  
       from order_details  
       join pizzas on pizzas.pizza_id = order_details.pizza_id)*100,2) as revenue  
from pizza_types join pizzas  
on pizza_types.pizza_type_id = pizzas.pizza_type_id  
join order_details  
on order_details.pizza_id = pizzas.pizza_id  
group by pizza_types.category order by revenue desc;
```

Result Grid					Filter F
	category	revenue			
▶	Classic	26.91			
	Supreme	25.46			
	Chicken	23.96			
	Veggie	23.68			

Analysis 12: Cumulative Revenue

Analyze the cumulative revenue generated over time.

- ```
select order_date,
sum(revenue) over(order by order_date) as cum_revenue
from
(select orders.order_date,
sum(order_details.quantity * pizzas.price) as revenue
from order_details join pizzas
on order_details.pizza_id = pizzas.pizza_id
join orders
on orders.order_id = order_details.order_id
group by orders.order_date) as sales;
```

| Result Grid |            |                    | Filter Rows: |
|-------------|------------|--------------------|--------------|
|             | order_date | cum_revenue        |              |
| ▶           | 2015-01-01 | 2713.8500000000004 |              |
|             | 2015-01-02 | 5445.75            |              |
|             | 2015-01-03 | 8108.15            |              |
|             | 2015-01-04 | 9863.6             |              |
|             | 2015-01-05 | 11929.55           |              |
| Result 2 x  |            |                    |              |

## Analysis 13: Top 3 by Revenue in Each Category

Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
• select category, name, revenue
 from
 (select category, name, revenue,
 rank() over(partition by category order by revenue desc) as Rnk
 from
 (select pizza_types.category, pizza_types.name,
 sum(order_details.quantity * pizzas.price) as revenue
 from pizza_types join pizzas
 on pizza_types.pizza_type_id = pizzas.pizza_type_id
 join order_details
 on order_details.pizza_id = pizzas.pizza_id
 group by pizza_types.category, pizza_types.name) as a) as b
 where Rnk <= 3;
```

| Result Grid |          |                              |                   |
|-------------|----------|------------------------------|-------------------|
|             |          | Filter Rows:                 | Export: W         |
|             | category | name                         | revenue           |
| ▶           | Chicken  | The Thai Chicken Pizza       | 43434.25          |
|             | Chicken  | The Barbecue Chicken Pizza   | 42768             |
|             | Chicken  | The California Chicken Pizza | 41409.5           |
|             | Classic  | The Classic Deluxe Pizza     | 38180.5           |
|             | Classic  | The Hawaiian Pizza           | 32273.25          |
|             | Classic  | The Pepperoni Pizza          | 30161.75          |
|             | Supreme  | The Spicy Italian Pizza      | 34831.25          |
|             | Supreme  | The Italian Supreme Pizza    | 33476.75          |
|             | Supreme  | The Sicilian Pizza           | 30940.5           |
|             | Veggie   | The Four Cheese Pizza        | 32265.70000000065 |
|             | Veggie   | The Mexicana Pizza           | 26780.75          |
|             | Veggie   | The Five Cheese Pizza        | 26066.5           |



# Key Findings

- Total Orders: 21,350
- Revenue: 817,860.05
- Large pizzas were ordered most frequently.
- Premium pizzas generated significant revenue.
- Peak demand occurs during lunch and evening.

# Business Recommendations

- Optimize inventory for popular pizzas.
- Increase staffing during peak hours.
- Promote low-selling categories.
- Bundle premium pizzas with beverages.

# Thank You

Thank you for viewing this project.

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